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Week 1-2: Introduction to Power System

Planning

PSM-104

M. Tech (EE)

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Comprehensive Class Notes

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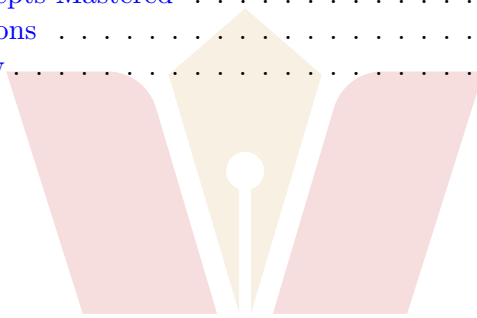
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1 Course Overview & Weekly Structure

The main aim of the present course is to provide comprehensive understanding of power system planning methodologies, reliability assessment techniques, and strategic decision-making frameworks essential for modern electrical engineers. This course encompasses both theoretical foundations and practical applications in contemporary power system development.

Definition 1.1 (Course Philosophy): *Power system planning represents the cornerstone of reliable electricity supply, requiring integration of technical, economic, environmental, and regulatory considerations to ensure sustainable energy infrastructure development.*

1.1 Weekly Learning Structure

Table 1: Power System Planning Course Structure - Week 1-2

Lecture	Topic	Duration	Learning Activities
1.1	Introduction to System Planning	50 min	Interactive case study discussion
1.2	Planning Objectives and Constraints	50 min	Problem-solving workshop
1.3	Short-term Planning Strategies	50 min	Load forecasting exercise
1.4	Medium-term Planning Approaches	50 min	Capacity planning simulation
1.5	Long-term Planning Methodologies	50 min	Strategic planning case study
1.6	Reactive Power Planning	50 min	MATLAB/OCTAVE simulation concepts
1.7	Planning Assessment and Review	50 min	Module assessment

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2 Introduction to Power System Planning

Power system planning constitutes the systematic process of determining the optimal expansion and operation of electrical power infrastructure to meet projected electricity demand while maintaining prescribed reliability levels and economic efficiency.

2.1 Historical Evolution of Power System Planning

The evolution of power system planning has progressed through distinct phases, each characterized by specific technological, economic, and regulatory paradigms.

2.1.1 Classical Planning Era (1920-1970)

During this foundational period, power system planning focused primarily on:

- **Centralized generation:** Large thermal and hydro power plants
- **Vertically integrated utilities:** Single entity controlling generation, transmission, and distribution
- **Load growth accommodation:** Simple extrapolation-based demand forecasting

- **Engineering-driven decisions:** Minimal economic optimization

Definition 2.1 (Classical Planning Equation): *The fundamental relationship governing classical planning was:*

$$C_{installed} = D_{peak} \times (1 + R_{margin}) \quad (1)$$

where typical reserve margins ranged from 15-20% to ensure adequate reliability.

2.1.2 Modern Planning Era (1970-2000)

The oil crises of the 1970s catalyzed significant changes in planning methodologies:

1. **Economic optimization:** Introduction of least-cost planning principles
2. **Environmental considerations:** Environmental impact assessments
3. **Demand-side management:** Load management and energy efficiency programs
4. **Probabilistic methods:** Reliability-based planning criteria

2.1.3 Contemporary Planning Era (2000-Present)

Current planning frameworks address unprecedented complexity:

- **Market deregulation:** Competitive electricity markets
- **Renewable integration:** Variable generation resources
- **Smart grid technologies:** Advanced monitoring and control systems
- **Climate change mitigation:** Carbon emission constraints
- **Cyber security:** Protection against digital threats

2.2 Fundamental Planning Paradigms

2.2.1 Deterministic vs. Probabilistic Planning

Definition 2.2 (Deterministic Planning): *Assumes fixed values for planning parameters, typically using worst-case scenarios. The planning criterion is expressed as:*

$$C_{required} = D_{maximum} + R_{deterministic} \quad (2)$$

Definition 2.3 (Probabilistic Planning): *Incorporates uncertainty through probability distributions. The reliability constraint becomes:*

$$P(\text{System Failure}) \leq \text{Acceptable Risk Level} \quad (3)$$

2.2.2 Contemporary Planning Challenges

The integration of variable renewable energy sources introduces new planning complexities:

$$\text{Effective Capacity} = \text{Nameplate Capacity} \times \text{Capacity Credit} \quad (4)$$

where capacity credit depends on:

- Resource availability correlation with peak demand
- Geographic diversity of renewable resources
- Energy storage system capabilities
- Demand response program effectiveness

Example 2.1 (Renewable Integration):

Consider a system with 1000 MW peak demand and 300 MW of wind generation:

$$\text{Wind Capacity Credit} = 0.3 \text{ (typical for wind)} \quad (5)$$

$$\text{Effective Wind Capacity} = 300 \times 0.3 = 90 \text{ MW} \quad (6)$$

$$\text{Additional Firm Capacity Required} = 1000 - 90 = 910 \text{ MW} \quad (7)$$

3 Planning Objectives and Constraints

Power system planning optimization requires balancing multiple, often conflicting objectives within a complex constraint framework.

3.1 Primary Planning Objectives

3.1.1 Economic Efficiency

The fundamental economic objective seeks to minimize total system cost:

$$\min \left\{ \sum_{t=1}^T \frac{1}{(1+r)^t} [\text{CAPEX}_t + \text{OPEX}_t + \text{ENVEX}_t] \right\} \quad (8)$$

where:

- CAPEX_t = Capital expenditure in year t
- OPEX_t = Operating expenditure in year t
- ENVEX_t = Environmental externality costs in year t
- r = Discount rate
- T = Planning horizon

Definition 3.1 (Net Present Value Calculation): *The economic evaluation employs net present value methodology:*

$$\text{NPV} = \sum_{t=0}^T \frac{\text{Cash Flow}_t}{(1+r)^t} \quad (9)$$

Typical utility discount rates range from 6-12% depending on regulatory framework and market conditions.

3.1.2 Reliability Assurance

Reliability objectives ensure adequate system performance under various operating conditions:

$$\text{LOLP} \leq \text{LOLP}_{\text{target}} \quad (10)$$

$$\text{EENS} \leq \text{EENS}_{\text{target}} \quad (11)$$

$$\text{SAIFI} \leq \text{SAIFI}_{\text{target}} \quad (12)$$

$$\text{SAIDI} \leq \text{SAIDI}_{\text{target}} \quad (13)$$

where:

- LOLP = Loss of Load Probability
- EENS = Expected Energy Not Served
- SAIFI = System Average Interruption Frequency Index
- SAIDI = System Average Interruption Duration Index

3.2 Planning Constraints Framework

3.2.1 Technical Constraints

Power system physics impose fundamental technical limitations:

Power Balance Constraint:

$$\sum_{i=1}^{N_g} P_{g,i} = \sum_{j=1}^{N_d} P_{d,j} + P_{\text{losses}} \quad (14)$$

Transmission Capacity Constraints:

$$|P_{\text{flow},k}| \leq P_{\text{capacity},k} \quad \forall k \in \text{transmission lines} \quad (15)$$

Generation Capacity Constraints:

$$P_{\min,i} \leq P_{g,i} \leq P_{\max,i} \quad \forall i \in \text{generators} \quad (16)$$

Voltage Constraints:

$$V_{\min} \leq |V_n| \leq V_{\max} \quad \forall n \in \text{buses} \quad (17)$$

Example 3.1 (Technical Constraint):

For a 230 kV transmission line with thermal rating 400 MVA:

$$\text{Power transfer limit} = 400 \text{ MVA} \quad (18)$$

$$\text{At unity power factor} = 400 \text{ MW} \quad (19)$$

$$\text{Voltage limits} = 230 \pm 10\% = 207 - 253 \text{ kV} \quad (20)$$

3.3 Multi-objective Optimization

Real-world planning problems require simultaneous optimization of multiple objectives. Common approaches include:

3.3.1 Weighted Sum Method

$$\min \{w_1 f_{cost} + w_2 f_{reliability} + w_3 f_{environment}\} \quad (21)$$

where w_i represents the relative importance weights.

Definition 3.2 (Planning Trade-offs): *Modern power system planning inherently involves trade-offs between cost, reliability, and environmental impact. Optimal solutions represent balanced compromises rather than absolute minima in any single dimension.*

4 Short-term Planning Strategies

Short-term planning encompasses operational planning and resource adequacy assessment over horizons typically ranging from one day to two years ahead.

4.1 Planning Horizon Classification

4.1.1 Operational Planning (Minutes to Hours)

Real-time and near-real-time operational decisions:

- **Economic dispatch:** Optimal generation scheduling
- **Unit commitment:** Generator on/off decisions
- **Load following:** Automatic generation control
- **Contingency response:** Emergency operating procedures

4.1.2 Short-term Strategic Planning (Months to 2 Years)

Resource adequacy and operational capability assessment:

- **Capacity assessment:** Peak demand vs. available generation
- **Transmission adequacy:** Transfer capability analysis
- **Resource procurement:** Capacity market participation
- **System performance:** Reliability metric evaluation

4.2 Load Forecasting Methodologies

Accurate load forecasting forms the foundation of short-term planning.

4.2.1 Time Series Analysis

Autoregressive Integrated Moving Average (ARIMA) Model:

$$\phi(B)(1 - B)^d X_t = \theta(B)\epsilon_t \quad (22)$$

where:

- $\phi(B)$ = Autoregressive operator
- $\theta(B)$ = Moving average operator
- B = Backshift operator
- d = Degree of differencing
- ϵ_t = White noise error term

4.2.2 Regression Analysis

Multiple Linear Regression Model:

$$P_{L,t} = \beta_0 + \beta_1 T_t + \beta_2 H_t + \beta_3 D_t + \beta_4 E_t + \epsilon_t \quad (23)$$

where:

- T_t = Temperature variable
- H_t = Humidity variable
- D_t = Day-of-week variable
- E_t = Economic activity indicator
- ϵ_t = Random error term

4.3 Resource Adequacy Assessment

4.3.1 Deterministic Criteria

Traditional planning reserves ensure adequate capacity:

$$\text{Planning Reserve Margin} = \frac{\text{Installed Capacity} - \text{Peak Demand}}{\text{Peak Demand}} \times 100\% \quad (24)$$

Typical reserve margins:

- **Generation reserves:** 15-20%
- **Transmission reserves:** 10-15%
- **Operating reserves:** 3-5%

4.3.2 Probabilistic Assessment

Loss of Load Probability Calculation:

$$\text{LOLP} = \sum_{k=0}^{\infty} P(\text{Available Capacity} = k) \times P(\text{Load} > k) \quad (25)$$

Definition 4.1 (Probabilistic Planning Advantage): Probabilistic methods provide more accurate risk assessment by explicitly modeling:

- Generation unit forced outage rates
- Load forecast uncertainty
- Transmission system contingencies
- Renewable resource variability

5 Medium-term Planning Approaches

Medium-term planning bridges the gap between operational planning and long-term strategic decisions, typically addressing timeframes from 1 to 5 years.

5.1 Planning Horizon Characteristics

Medium-term planning focuses on tactical decisions that support strategic objectives while maintaining operational flexibility.

5.1.1 Key Planning Elements

- **Capacity resource assessment:** Evaluation of existing generation adequacy
- **Transmission reinforcement:** Identification of network bottlenecks
- **Load growth accommodation:** Demand forecast validation and adjustment
- **Technology transition:** Retirement and replacement planning
- **Environmental compliance:** Regulatory requirement implementation

5.2 Capacity Planning Simulation

Medium-term capacity planning employs sophisticated simulation models to evaluate resource adequacy under various scenarios.

5.2.1 Planning Reserve Margin Analysis

The fundamental capacity planning equation ensures adequate reserves:

$$\text{Required Capacity} = \text{Peak Demand} \times (1 + \text{Planning Reserve Margin}) + \text{Transmission Losses} \quad (26)$$

Example 5.1 (Capacity Planning Simulation):

Consider a system with the following characteristics:

- Current peak demand: 8,500 MW
- Annual growth rate: 3.2%
- Planning reserve margin: 17%
- Transmission losses: 4%

For Year 3 planning:

$$\text{Projected Peak Demand} = 8,500 \times (1.032)^3 = 9,321 \text{ MW} \quad (27)$$

$$\text{Required Capacity} = 9,321 \times 1.17 \times 1.04 = 11,353 \text{ MW} \quad (28)$$

6 Long-term Planning Methodologies

Long-term planning addresses strategic infrastructure development over horizons extending from 10 to 30 years, focusing on major technology transitions and system transformation.

6.1 Strategic Planning Framework

Long-term methodologies must address fundamental uncertainties in technology costs, environmental regulations, and energy demand patterns.

6.1.1 Scenario-Based Planning

Multiple scenario development captures uncertainty ranges:

$$\text{Expected NPV} = \sum_{s=1}^S \pi_s \times \text{NPV}_s \quad (29)$$

where π_s represents scenario probability and NPV_s is the net present value under scenario s .

Common Scenario Dimensions:

- **Economic growth:** High/medium/low GDP growth scenarios
- **Energy prices:** Fossil fuel cost trajectories
- **Technology advancement:** Renewable cost reduction rates
- **Policy evolution:** Carbon pricing and environmental regulations
- **Demand patterns:** Electrification of transportation and heating

Example 6.1 (Strategic Planning Case Study):

A utility planning 20-year transformation must consider:

- Coal plant retirement schedule (environmental compliance)
- Natural gas transition strategy (bridge technology)
- Renewable energy deployment (cost competitiveness)
- Energy storage integration (grid flexibility)
- Transmission network expansion (renewable integration)

6.2 Generation Planning Models

6.2.1 Least-Cost Planning Framework

The optimization objective minimizes system-wide costs:

$$\min \sum_{t=1}^T \sum_{i=1}^N \frac{1}{(1+r)^t} [CC_i x_{i,t} + FC_i P_{i,t} + VC_i E_{i,t}] \quad (30)$$

where:

- CC_i = Capital cost of technology i
- FC_i = Fixed O&M cost of technology i
- VC_i = Variable O&M cost of technology i
- $x_{i,t}$ = Capacity additions of technology i in year t
- $P_{i,t}$ = Installed capacity of technology i in year t
- $E_{i,t}$ = Energy production from technology i in year t

6.2.2 Technology Characterization

Technology	Capital Cost (\$/kW)	Fixed O&M (\$/kW-yr)	Variable O&M (\$/MWh)	Capacity Factor (%)
Coal Steam	3,500	45	30	85
CCGT	1,200	15	7	60
Solar PV	1,400	20	0	25
Wind Onshore	1,600	30	0	35
Nuclear	6,000	95	12	90
Hydro	2,800	40	5	50

7 Reactive Power Planning

Reactive power planning ensures voltage stability and power quality throughout the transmission and distribution network while minimizing system losses.

7.1 Reactive Power Fundamentals

Reactive power planning addresses voltage control requirements and power factor optimization across different system operating conditions.

7.1.1 Voltage Control Hierarchy

The voltage control system operates at multiple levels:

1. **Primary control:** Generator automatic voltage regulators (seconds)
2. **Secondary control:** Load tap changers and switched capacitors (minutes)
3. **Tertiary control:** Coordinated voltage/VAR optimization (hours)

7.2 Reactive Power Planning Analysis

7.2.1 Voltage Stability Margin

The voltage stability margin quantifies system proximity to voltage collapse:

$$\text{VSM} = \frac{P_{\max} - P_{\text{operating}}}{P_{\max}} \times 100\% \quad (31)$$

where $P_{\max}$ is the maximum loadability and $P_{\text{operating}}$ is the current operating point.

Example 7.1 (MATLAB/OCTAVE Simulation Concepts):

Key simulation components for reactive power planning:

- **Load flow analysis:** Steady-state voltage profile calculation
- **Continuation power flow:** Voltage stability margin assessment
- **Optimal power flow:** Economic reactive power dispatch
- **Time-series simulation:** Daily/seasonal reactive power requirements

MATLAB/OCTAVE toolboxes: Power System Toolbox, MATPOWER, PSAT

8 Planning Assessment and Review

The planning assessment process evaluates the effectiveness of planning methodologies and validates planning assumptions through systematic review procedures.

8.1 Assessment Framework

Planning assessment encompasses multiple evaluation criteria to ensure comprehensive performance measurement.

8.1.1 Planning Performance Metrics

Reliability Metrics:

- **Loss of Load Probability (LOLP):** Days per year with supply shortfall
- **Expected Energy Not Served (EENS):** Annual energy shortfall
- **System reliability indices:** SAIFI, SAIDI, CAIDI measurements

Economic Metrics:

- **Total system cost:** Capital + operating + environmental costs
- **Customer cost impact:** Rate increase projections
- **Benefit-cost ratio:** Investment justification analysis

8.2 Module Assessment Methodology

8.2.1 Comprehensive Evaluation Framework

The module assessment integrates theoretical understanding with practical application:

Definition 8.1 (Week 1-2 Assessment Components):

- **Class Discussion (40%):** Planning case study analysis and peer interaction
- **Programming Assignment (45%):** Load forecasting algorithm implementation
- **Participation (15%):** Active engagement in learning activities

9 Reliability Fundamentals

Power system reliability quantifies the ability to provide continuous, adequate electricity supply under normal and contingency conditions.

9.1 Reliability Definitions and Metrics

9.1.1 Fundamental Reliability Indices

System Average Interruption Frequency Index:

$$\text{SAIFI} = \frac{\sum \text{Customer Interruptions}}{\text{Total Customers Served}} \quad (32)$$

System Average Interruption Duration Index:

$$\text{SAIDI} = \frac{\sum \text{Customer Interruption Durations}}{\text{Total Customers Served}} \quad (33)$$

Customer Average Interruption Duration Index:

$$\text{CAIDI} = \frac{\text{SAIDI}}{\text{SAIFI}} = \frac{\sum \text{Customer Interruption Durations}}{\sum \text{Customer Interruptions}} \quad (34)$$

Example 9.1 (Reliability Index Calculation):

For a utility serving 100,000 customers with annual data:

- Total interruptions: 150,000
- Total interruption hours: 450,000

$$\text{SAIFI} = \frac{150,000}{100,000} = 1.5 \text{ interruptions/customer} \quad (35)$$

$$\text{SAIDI} = \frac{450,000}{100,000} = 4.5 \text{ hours/customer} \quad (36)$$

$$\text{CAIDI} = \frac{4.5}{1.5} = 3.0 \text{ hours/interruption} \quad (37)$$

9.2 Probabilistic Reliability Assessment**9.2.1 Component Reliability Modeling**

Individual component reliability follows exponential distribution:

$$R(t) = e^{-\lambda t} \quad (38)$$

where λ is the failure rate (failures per unit time).

Component Availability:

$$A = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} = \frac{1/\lambda}{1/\lambda + r} \quad (39)$$

where:

- MTTF = Mean Time To Failure
- MTTR = Mean Time To Repair
- r = Repair rate

10 Planning Criteria & Standards

Planning criteria and standards establish the technical and regulatory framework governing power system development and operation.

10.1 Regulatory Framework

10.1.1 International Standards

IEEE Standards:

- IEEE 762: Reliability Data for Generating Units
- IEEE 859: Standard Terms for Reporting System Reliability
- IEEE 1366: Guide for Electric Power Distribution Reliability Indices

10.2 Technical Planning Criteria

10.2.1 Adequacy Criteria

Generation Adequacy:

$$\text{Installed Capacity} \geq \text{Peak Demand} \times (1 + \text{Reserve Margin}) \quad (40)$$

Typical reserve margins by region:

- ERCOT (Texas): 13.75%
- PJM (Mid-Atlantic): 15.3%
- CAISO (California): 15.0%
- WECC (Western US): 12.0%

10.3 Economic Planning Criteria

10.3.1 Benefit-Cost Analysis

Investment justification requires positive net present value:

$$\text{NPV} = \sum_{t=1}^T \frac{\text{Benefits}_t - \text{Costs}_t}{(1+r)^t} > 0 \quad (41)$$

Example 10.1 (Economic Evaluation):

For a transmission line project costing \$100M with benefits:

- Annual production cost savings: \$5M
- Annual reliability improvements: \$2M
- Project life: 40 years
- Discount rate: 7%

$$\text{Annual Benefits} = \$5M + \$2M = \$7M \quad (42)$$

$$\text{NPV} = -\$100M + \$7M \times \frac{1 - (1.07)^{-40}}{0.07} \quad (43)$$

$$= -\$100M + \$7M \times 13.33 = \$93.3M > 0 \quad (44)$$

Project is economically justified.

11 Summary and Key Learning Outcomes

11.1 Fundamental Concepts Mastered

Through the comprehensive coverage of Week 1-2 material, students have developed understanding of:

1. **Planning fundamentals:** Historical evolution and contemporary challenges in power system planning
2. **Objective frameworks:** Multi-objective optimization balancing economic, reliability, and environmental considerations
3. **Constraint management:** Technical, regulatory, and financial limitations in planning decisions
4. **Short-term strategies:** Load forecasting, resource adequacy, and operational planning methodologies
5. **Reliability quantification:** Mathematical frameworks for system reliability assessment
6. **Regulatory compliance:** Planning criteria and standards governing system development

11.2 Practical Applications

The theoretical foundations enable practical applications in:

- **System planning studies:** Conducting comprehensive planning analyses for real utilities
- **Investment evaluation:** Economic assessment of generation and transmission projects
- **Reliability assessment:** Quantitative evaluation of system reliability performance
- **Regulatory compliance:** Ensuring planning studies meet industry standards
- **Strategic decision-making:** Supporting long-term resource planning decisions

Definition 11.1 (Course Learning Progression): *The foundational knowledge developed in these initial weeks provides the essential framework for advanced topics including renewable integration planning, smart grid deployment strategies, and resilience enhancement methodologies to be covered in subsequent weeks.*

11.3 Next Week Preview

Week 3-4 will build upon these foundations to explore:

- **Advanced optimization techniques:** Stochastic and robust optimization methods
- **Renewable integration:** Variable generation planning and grid flexibility requirements
- **Market mechanisms:** Capacity markets and reliability procurement
- **Resilience planning:** Extreme event preparation and system hardening strategies

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